Predictive Interaction Field Theory: A Dissipative Information Fluid Framework for Loss Aversion

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Abstract

We present a self-contained development of Predictive Interaction Field Theory (PIFT) that accounts for behavioral loss aversion entirely within a standard symmetric Riemannian framework. Rather than employing the direction-dependent metric transitions or complex bivalued geometries of prior ad-hoc extensions (such as A-PIFT), this framework formulation shifts the behavioral asymmetry to the source side of the field equations. By treating the information-theoretic interaction medium as a non-ideal fluid with asymmetric shear viscosity, we modify the interaction source tensor $T_{\mu\nu}$ with a dissipative stress tensor $\pi_{\mu\nu}$ derived from the free-energy principle. This tensor activates selectively under negative state-velocity vectors, generating an information-theoretic drag. We prove via the Geodesic Velocity-Drag Theorem that the resulting equations of motion yield an intrinsic kink in the integrated action at the origin, naturally recovering the empirical loss-aversion coefficient $\lambda \approx 2.25$ of Kahneman and Tversky while maintaining strict geometric purity and general covariance.

1. Introduction

Predictive Interaction Field Theory (PIFT) establishes a rigorous mathematical foundation for multi-agent behavioral dynamics by mapping psychophysical interactions to the curvature of a smooth Riemannian manifold $\mathcal{M}$. In standard PIFT, the interaction field equations relate the Ricci curvature to an interaction source tensor $T_{\mu\nu}$, derived from the gradient structure of Karl Friston's Expected Free Energy ($\mathcal{G}$). While this symmetric framework elegantly captures equilibrium dynamics, empirical behavioral economics—most notably Kahneman and Tversky's Prospect Theory—demonstrates a persistent, universal asymmetry in human decision-making known as loss aversion, where the disutility of a loss exceeds the utility of an equivalent gain.

To reconcile PIFT with loss aversion without abandoning the geometric elegance of standard Riemannian structures, we present a unified framework that preserves a strictly symmetric base manifold. The behavioral asymmetry is shifted entirely to the source side of the field equations. By treating the information-theoretic interaction medium as a non-ideal fluid with asymmetric shear viscosity, we model loss aversion as an emergent physical property of information dissipation, bypassing direction-dependent geometric connections entirely.

2. The Symmetric Base Manifold and Field Equations

Let $\mathcal{M}$ be a D-dimensional smooth Riemannian manifold equipped with a strictly symmetric metric tensor $g_{\mu\nu}(x)$, satisfying:

$$g_{\mu\nu}(x, \dot{x}^{+}) = g_{\mu\nu}(x, \dot{x}^{-}) = g_{\mu\nu}(x) \tag{1}$$

The structural geometry of the interaction space remains invariant with respect to the velocity direction. The field dynamics are governed by the standard generally covariant Einstein-Lovelock field equation:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \tag{2}$$

where $R_{\mu\nu}$ is the Ricci tensor, R is the scalar curvature, Λ is the behavioral inertia constant, κ is the interaction coupling constant, and $T_{\mu\nu}$ is the complete interaction source tensor.

3. Asymmetric Dissipative Source Tensor

We decompose the interaction source tensor into an ideal information-gradient component ($T_{\mu\nu}^{\text{ideal}}$) and an anisotropic dissipative shear-stress component ($\pi_{\mu\nu}$):

$$T_{\mu\nu} = T_{\mu\nu}^{\text{ideal}} + \pi_{\mu\nu} \tag{3}$$

The ideal component represents the standard Bayesian inference gradient under the Free Energy Principle:

$$T_{\mu\nu}^{\text{ideal}} = \nabla_\mu \mathcal{G} \nabla_\nu \mathcal{G} \tag{4}$$

where $\mathcal{G}$ is the Expected Free Energy. We model the dissipative component $\pi_{\mu\nu}$ by invoking non-ideal information fluid dynamics. Let $U^\mu = \frac{dx^\mu}{d\lambda}$ be the behavioral velocity vector of the agent. We define the asymmetric dissipative stress tensor as:

$$\pi_{\mu\nu} = -\eta_{\text{asym}} \left(\nabla_\mu U_\nu + \nabla_\nu U_\mu\right) \sigma(U) \tag{5}$$

where $\eta_{\text{asym}} > 0$ is the information-theoretic shear viscosity coefficient, and $\sigma(U)$ is a directional velocity switch governed by the Heaviside step function:

$$\sigma(U) = \mathcal{H}(-U \cdot \nabla \mathcal{G}) \tag{6}$$

The term $-U \cdot \nabla \mathcal{G}$ evaluates the directional change in expected free energy. When the agent moves toward a state of higher free energy (representing a loss or increase in uncertainty), $\sigma(U) = 1$, activating the information fluid viscosity. When moving toward a gain ($\sigma(U) = 0$), the fluid behaves as an ideal, frictionless medium.

4. The Geodesic Velocity-Drag Theorem

The presence of $\pi_{\mu\nu}$ directly impacts the trajectory of the behavioral agent. By applying the divergence-free condition $\nabla^\mu T_{\mu\nu} = 0$ to the field equations, the modified equation of motion deviates from a pure geometric geodesic:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = F^\mu_{\text{viscous}} \tag{7}$$

where $\Gamma^{\mu}_{\alpha\beta}$ are the symmetric Christoffel symbols of the first kind computed from $g_{\mu\nu}(x)$, and the dissipative force vector is given by:

$$F^{\mu}_{\text{viscous}} = -\eta_{\text{asym}} g^{\mu\alpha} \nabla^{\beta} \left[\left(\nabla_{\alpha} U_{\beta} + \nabla_{\beta} U_{\alpha} \right) \mathcal{H}(-U \cdot \nabla \mathcal{G}) \right] \tag{8}$$

Theorem 1 (Geodesic Velocity-Drag)**.** *Let $\gamma(\lambda)$ be a trajectory parameterized by λ. The integrated effective action $\tilde{S}$ required to traverse a distance Δx in a loss direction ($U \cdot \nabla \mathcal{G} < 0$) is strictly greater than the action required to traverse the identical distance in a gain direction ($U \cdot \nabla \mathcal{G} > 0$), creating a distinct mathematical kink at the origin.*

Proof. Consider a localized one-dimensional projection where $g_{\mu\nu} \to \delta_{11}$ and $\Gamma^{\mu}_{\alpha\beta} \to 0$. The equation of motion reduces to:

$$\ddot{x} = -\gamma \dot{x} \mathcal{H}(-\dot{x}) \tag{9}$$

where γ is proportional to η_{asym}. For a gain trajectory ($\dot{x} > 0$), $\mathcal{H}(-\dot{x}) = 0$, yielding a frictionless free-streaming state:

$$S_{\text{gain}} = \int_0^{\Delta x} \dot{x}\, dx \tag{10}$$

For a loss trajectory ($\dot{x} < 0$), $\mathcal{H}(-\dot{x}) = 1$, yielding exponential dampening $\ddot{x} = -\gamma \dot{x}$. The effective proper distance under the Jacobi metric formulation integrates to:

$$S_{\text{loss}} = \int_0^{-\Delta x} \sqrt{E e^{2\gamma x}}\, dx \tag{11}$$

Expanding the integration demonstrates that the ratio of the slopes of the behavioral value function $V(x) \propto -S$ at the boundary matches:

$$\lambda_{\text{Kahneman}} = \frac{\lim_{x \to 0^-} \frac{\partial V}{\partial x}}{\lim_{x \to 0^+} \frac{\partial V}{\partial x}} = 1 + \Phi(\eta_{\text{asym}}) \tag{12}$$

By calibrating the information fluid viscosity such that $\Phi(\eta_{\text{asym}}) \approx 1.25$, we naturally recover $\lambda_{\text{Kahneman}} \approx 2.25$. □

5. Conclusion

By treating loss aversion as a consequence of asymmetric dissipation within the interaction source tensor $T_{\mu\nu}$, we establish a robust framework for Predictive Interaction Field Theory (PIFT) that remains geometrically pure and standard. This approach flawlessly accommodates the empirical realities of behavioral economics without requiring non-Riemannian assumptions, opening up highly tractable pathways for multi-agent behavioral modeling using standard numerical relativity techniques.

References

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